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1. Permanent Magnet Moving Coil (PMMC) Instrument

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Construction:

Permanent Magnet → Provides a strong uniform magnetic field

Moving Coil → Placed between magnet poles, wound on aluminum frame

Soft Iron Core → Improves magnetic field strength

Spring (Control Spring) → Provides controlling torque

Pointer & Scale → Shows reading

Damping → Eddy current damping (due to aluminum frame)

Principle:

Works on Lorentz force principle:
A current-carrying conductor placed in a magnetic field experiences a force.

Operation:

When DC current flows through the coil → magnetic field is produced.

Interaction with permanent magnetic field → force acts on coil.

Coil rotates → pointer moves on scale.

Spring provides opposing torque → system stabilizes.

Important Result:

 Deflection is directly proportional to current
→ Uniform scale (very important point for exams)

Advantages:

High accuracy

Uniform scale

Low power consumption

Good damping

Disadvantages:

Works only with DC (Direct Current)

Cannot measure AC directly

Applications:

DC ammeters

DC voltmeters

Galvanometers

4. Moving Iron Instruments (Attraction & Repulsion)

Moving Iron (MI) instruments are commonly used for measuring AC and DC currents and voltages. They work on the principle that a piece of iron gets magnetized when placed in a magnetic field and experiences a mechanical force. MI instruments are of two types:

Repulsion type and Attraction type, each with slightly different operation. Here’s a simple explanation:

1. Repulsion Type Moving Iron Instruments

Construction & Principle:

Contains two iron pieces, one fixed and one movable.

When current flows through the coil, both iron pieces are magnetized with like poles facing each other.

Like poles repel each other.

Function:

The movable iron piece rotates due to magnetic repulsion.

Rotation is opposed by a spring, which gives a deflection proportional to current/voltage.

Used to measure AC or DC.

Characteristics:

Better accuracy at low currents.

Less friction because movable part is light.

2. Attraction Type Moving Iron Instruments

Construction & Principle:

Contains one fixed coil and a movable iron vane/piece.

The coil is energized by the measured current.

The iron is attracted towards the coil due to magnetization.

Function:

The movable iron moves toward the coil.

A spring provides a counter torque, so the deflection is proportional to current/voltage.

Can also measure AC or DC.

Characteristics:

Simpler in construction.

Generally used for low-cost instruments.

Key Difference Between Repulsion & Attraction MI Instruments

Feature

Repulsion Type

Attraction Type

Force